



WHITEPAPER

TeTe

**Peer-to-peer skill challenges on Nimiq. Stake
on what you're good at — never on chance.**

teteonnimiq.site

September 2026

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01 Abstract

TeTe is a peer-to-peer skill-challenge platform built as a Nimiq Pay Mini App. Two players agree a challenge, put up an equal stake in NIM, compete at something they are actually good at, and the stake goes to the winner — decided entirely by skill, never by chance.

TeTe never touches a private key or a seed phrase; every wallet action is approved by the player inside Nimiq Pay. Stakes are held by a disclosed, bounded custodial escrow — not because that was the ideal design, but because it is the only shape the Mini App provider's transaction surface allows today, and TeTe says so plainly rather than implying trustless custody it cannot deliver. Deposits are verified against the treasury's own on-chain transaction history rather than taken on a player's word, disputes are held rather than auto-resolved, and every screen refuses to show a number it cannot back with a real check.

Alongside staked play, TeTe ships a free arcade of eight original skill games that credit real, withdrawable NIM rewards for how well you play — no stake required, nothing decided by luck. This document describes what TeTe is, how a challenge actually moves money, the security and trust model behind it, and what is still ahead.

02 The problem

Betting a friend you'll win the next match is one of the oldest social contracts there is — and one of the least reliable. In practice, it breaks down in three predictable ways:

No neutral place to put the money

Someone has to hold both stakes until the result is known. That is either one of the two players — who now has an incentive to dispute a result they lost — or a messaging thread full of "I'll send it after," which quietly turns a bet into an unenforceable IOU the moment either side is less than fully honest.

Gambling platforms solve the wrong problem

Existing staking and betting apps are built around games of chance — dice, cards, slots, sportsbook odds. They are regulated as gambling for exactly that reason, and they have no real mechanism for "I am better at this specific game than you are, let's find out" — a claim that is provably true or false by playing, not by a random number generator.

Custodial apps ask for blind trust

Where a centralized app *does* hold stakes for a skill contest, a player is typically asked to trust that the operator received the deposit, will honor the result, and will not simply keep the money — with no way to check any of it. The custody itself is not always the problem; the opacity around it is.

03 The TeTe solution

TeTe answers each of these directly, rather than routing around them.

Skill

ONLY — NEVER
CHANCE

Verified

DEPOSITS, NOT
TRUSTED CLAIMS

Disclosed

CUSTODY, STATED
PLAINLY

Bounded

PAYOUT CEILING
PER TRANSACTION

A neutral place to put the money

Both players send their stake to a single treasury address that neither of them controls. Funding is confirmed by the server independently scanning the treasury's own transaction history for a matching payment — never by trusting what a wallet's send call reports, and never by trusting a player's claim that they paid.

Only skill, stated as a rule

Every format TeTe offers — esports titles, chess, trivia, a custom contest, or the built-in arcade — is decided by what a player actually did. Nothing in the product resolves on a random outcome, and nothing ever will; it is the one rule the roadmap will not trade away for growth.

Custody, disclosed rather than implied

TeTe's escrow is custodial, and it says so in the product itself, not just in this document. That trade-off exists because the Nimiq Pay Mini App provider can create basic and staking transactions on a player's behalf, but nothing that would let TeTe construct a trustless hash-timelock contract for them — so custody is the only shape that works through the provider today. What backs it: a payout ceiling on every single transaction, verification against the chain rather than a client's word, and every privileged action gated behind a signature TeTe cannot forge. Section 09 covers this in full.

"Decided by skill. Verified, not trusted. Disclosed, not implied."

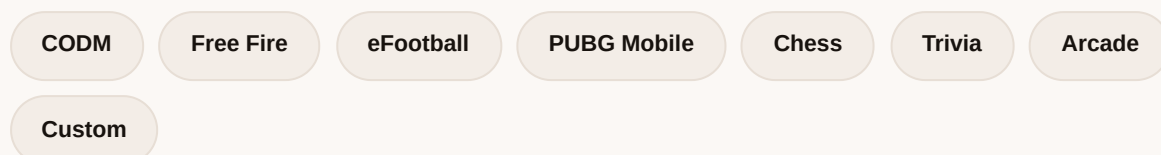
04 Product overview

TeTe runs as a Mini App inside Nimiq Pay — a lightweight web app loaded in the wallet's own WebView, with no separate install and direct access to the connected Nimiq and EVM wallets, each mediated by Nimiq Pay's own confirmation dialogs.

Five surfaces

SCREEN	WHAT IT DOES
Home	Stats, quick-create arenas, an arcade shortcut, live matches
Battles	Every challenge you're part of — drafts, awaiting stakes, live, settled
Create	The challenge builder — format, stake, opponent, share link
Arcade	Eight free skill games, sound effects, character skins, real rewards
Wallet / Profile / Leaderboard	Earnings, withdraw, tip, identity, and honest season standings

Eight challenge formats



A challenge can be aimed at a specific opponent by @username — resolved through a small public directory, so no address ever has to be typed twice — or posted open, for the first player to accept it. Terms travel in a shareable link that opens directly inside Nimiq Pay for whoever receives it.

05 How a challenge works

Every challenge moves through a small, explicit state machine. Nothing not listed below is a legal transition — a malformed or replayed request cannot walk a challenge into a state that pays out.

STATE	MEANING
open	Posted, waiting for someone to accept. Nothing funded.
accepted	An opponent joined. Both sides now owe their stake.
partly_funded	One side's stake confirmed on chain, the other outstanding.
funded	Both stakes confirmed. The match is live.
reported	At least one result reported, waiting for agreement.
disputed	Reports disagree. Held for resolution — nothing paid automatically.
settled	Paid to the winner. Terminal.
refunded	Called off before both sides funded, or a resolved dispute. Stakes returned.
expired	Never funded, past its deadline. Terminal, nothing owed.

Funding: verified, never trusted

A player claiming to have sent a stake proves nothing on its own. TeTe's server independently scans the treasury address's own transaction history for a confirmed payment matching the sender, the amount, and the challenge's reference tag before a challenge moves to funded. A dedicated diagnostic endpoint exists for exactly the case where a player insists they paid and the record disagrees — it shows what the treasury actually received, in every readable form, so a mismatched amount or a missing memo is visible rather than argued about.

Settlement and disputes

Once a match is played — off the app, on whatever platform the format is actually played on — both players report who won. Agreement pays the pot straight to the winner, waited-for on chain before TeTe reports success. Disagreement holds the challenge as disputed and pays nobody automatically: choosing a winner from contradictory claims is precisely the decision that needs a human, not a coin flip. Either side can offer to call a disputed match off and take their own stake back, but it takes *both* sides agreeing — otherwise a losing player could dispute every honest result and simply wait it out. If the two players cannot agree at all, a human operator — gated behind a bearer token held to the same standard as the treasury's own passphrase — can arbitrate, choosing only between the two outcomes the challenge ever allows: the pot to one side, or both stakes refunded.

Every write is signed

Posting, accepting, reporting, voiding, withdrawing, tipping and claiming a username are each authorized by a signature over that exact action and a timestamp, valid for five minutes. The server checks both that the signature verifies *and* that the address it derives to is the address being claimed — checking only the first would let anyone sign with their own key while claiming somebody else's identity.

06 The arcade

Not every session is worth staking on, so TeTe ships a free arcade alongside the staked side of the product — eight original games, each built in a familiar arcade *genre* rather than staked on anyone else's characters, art, or levels.

Crossing

Road-crossing. Hop the traffic, do not stop.

Drift

One-touch driving. Hold to steer, release to fall.

Slice

Swipe-to-slice. Chain combos, dodge the bombs.

Invasion

Fixed shooter. The formation speeds up as it thins.

Rush

Three-lane runner. Swipe to switch, jump, or roll.

Pitch

Top-down free kicks. Drag back, bend it round the wall.

Overheat

Physics trials. Manage engine heat and landings.

Alley

Street brawler. What's lying around decides the fight.

Every round is decided entirely by the player's own input — nothing pays out on chance here either. Each game plays a short synthesized sound effect on its core actions, ends a run with either a "You Lose" or a celebratory "New High Score!" moment, and lets a player pick from five character skins that carry across the games with a visible on-screen character.

07 Rewards and the PnL card

The arcade and a daily check-in credit real NIM to a server-side reward ledger as you play — the number on screen is a display copy of what the server has actually credited, never the other way around.

Withdraw

Pays exactly the server-tracked balance — never a client-supplied number — from a floor of 25 NIM, as a real signed transaction the withdrawal screen waits on before confirming.

Tip

Moves value directly between two players' ledgers by username — no on-chain transaction, no fee, no minimum — so a balance below the withdrawal floor is still worth something to its owner.

Activity

A public, server-recorded history per address, so a tip somebody else sent shows up on any device a player opens TeTe from — not only the one that received it.

The PnL card

Win a staked challenge and TeTe offers a downloadable image of the result — both players' real faces (a chosen photo or a deterministic generated avatar, whichever a player has actually set), handles, the format, the pot, the stake, and the moment it settled. It is drawn on a canvas from the challenge record itself at the moment of download rather than rendered once and stored, so a card downloaded weeks later looks exactly like one downloaded the instant the pot landed — there is nothing that can go stale, because nothing is cached.

08 Architecture

TeTe is a Next.js application. Blockchain interaction is kept strictly out of application logic: the client renders state and requests transitions, and the server is the only writer.

LAYER	TECHNOLOGY	ROLE
Wallet access	@nimiq/mini-app-sdk	Waits for Nimiq Pay to inject the Nimiq and EVM providers; TeTe never constructs one itself
Application	Next.js 16, React 19	Server route handlers hold every secret; nothing privileged runs in the client
Durable store	Redis-compatible KV	Challenge records, the username directory, reward ledgers, activity history
Treasury	Albatross RPC node	Holds the custodial escrow wallet, verifies deposits, signs payouts
EVM reads	viem	USDT balance and chain-switching support, ABI-encoded via a library rather than by hand

Fails closed

Without a configured durable store and treasury, TeTe runs in a local-only mode: challenges save as unfunded local drafts, and every screen that would need the backend says so plainly rather than offering an action it cannot complete. A single status endpoint exposes exactly two booleans — whether a store and a treasury are configured — and nothing else, so the client only ever offers what the deployment can actually do.

One EVM address, four verified chains

Nimiq Pay derives a single EVM wallet from the user's entropy and uses the same address on every supported chain. TeTe reads USDT balances on Polygon, Ethereum, Arbitrum One and Optimism against contract addresses published in Nimiq's own documentation, and reports the path as unavailable on any chain it cannot verify rather than guessing an address.

09 Security model

KEYS

TeTe never sees a private key or a seed phrase

It cannot: the WebView is sandboxed and Nimiq Pay mediates every wallet operation through its own native confirmation dialog. There is no code path in TeTe that could bypass this even if it wanted to.

The treasury is different, and stated plainly

It is a custodial hot wallet, unlocked on a server-controlled node by a passphrase held as sensitively as any credential in the system. It exists because the Mini App provider cannot build a trustless escrow primitive on a player's behalf. Its blast radius is bounded on purpose: a hard ceiling caps any single payout, every payout is either a verified settlement or an operator-arbitrated dispute — never a client-chosen amount — and the admin token that can touch the reward ledger is guarded at the same trust level as the treasury passphrase itself.

Every privileged write is signed and address-checked

A signature over the specific action and a timestamp, checked for validity *and* that the signer's derived address matches the identity being claimed — not taken on a player's word either way.

Minimal disclosure by default

The public status endpoint exposes only whether the backend is configured, never a URL, address, or token. Connection state lives in memory only — no cookie, no client-side analytics, and a device identifier is never requested at onboarding, only where anti-spam work would eventually need it.

10 Design philosophy: verified, never trusted

One rule shows up in nearly every decision TeTe makes: never show a number, a state, or a result that has not actually been checked.

- **A funded stake is a fact TeTe checked**, not a report a wallet handed back — a send call resolving successfully is not treated as proof of a payment, because it has been observed to resolve with nothing ever reaching the escrow.
- **A balance TeTe cannot verify is shown as unavailable** — never a zero, and never a dash styled to look like a number.
- **An empty leaderboard stays empty**, with the exact rules for how a rank will be earned, rather than sample opponents invented to look populated.
- **A disputed result is never resolved by a timer or a coin flip** — only by agreement between the two players, or a human who actually looked.
- **A reward ledger is the server's number**, and a withdrawal pays exactly that — never a figure the client supplies.

This is also why the custodial escrow trade-off is stated outright rather than left implicit: a product that is careful never to fabricate a balance has no business being vague about who is actually holding a stake while a match is live.

11 Roadmap

NEXT	WHAT'S INVOLVED
USDT escrow	The stake currency picker already offers it; on-chain funding verification and payout on the EVM side do not exist yet, and need a gas-awareness story of their own.
Reputation and ranking	The leaderboard's rules already exist; computing a rank from settled challenges does not yet.
Trustless escrow research	Custody is a disclosed trade-off, not a preference. Moving off it needs either a provider primitive the Mini App SDK does not expose today, or a different settlement design entirely.
Physical-device verification	Verified today against the documented API surface and a production build; not yet run inside Nimiq Pay on real hardware.
Nimiq Pay catalogue submission	A separate, GitHub-based process against the catalogue's own open-source repository, so the documented in-app deeplink starts resolving to TeTe directly.

One constraint does not move with any of these: only skill-based formats, ever. No games of chance.

12 Why Nimiq

TeTe is built as a Nimiq Pay Mini App rather than a standalone wallet integration for a specific reason: Nimiq Pay already puts a browser-based wallet in front of a mobile-first audience, with account access mediated entirely through its own native confirmation flow. That is exactly the trust boundary a peer-to-peer staking product needs — TeTe gets to build the challenge and escrow logic without ever being in a position to ask for, or need, a key.

Nimiq's own chain gives the NIM side of the product a fast, low-fee settlement layer for the transactions that actually matter here — a stake going in, a pot going out — while Nimiq Pay's EVM wallet support opens the same product up to USDT-denominated play on the chains real liquidity already sits on.

13 Disclaimer

TeTe has no token, coin, or tradable asset of its own, and this document describes a product, not an investment or a securities offering. Every challenge on TeTe is settled entirely by the players' own skill; TeTe does not offer, and will not add, any format decided by chance, and nothing here should be read as gambling in any form. Escrow custody is real, disclosed, and bounded as described in Sections 05 and 09 — using TeTe means trusting that custody model, the same way using any custodial service means trusting its operator, and players should weigh that trade-off for themselves before staking anything they are not prepared to lose to a dispute, an operator error, or a failure of the infrastructure TeTe depends on. Reward rates, thresholds, and supported chains are implementation details that may change as the product develops. This document reflects TeTe as shipped at the time of writing and is provided for informational purposes only.

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A Nimiq Pay Mini App · Stake on skill. Play free in the arcade.